Colter Snyder

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SUMMARY

Full-stack software engineer building probabilistic nuclear weapons effects simulation platforms for defense customers in CUI/ITAR-controlled environments, with Angular/Tauri front ends, Rust/Python back ends, and Fortran-to-Rust model ports. M.S. in Computer Science (3.97 GPA); pursuing an M.Eng. in Nuclear Engineering. Former adjunct instructor who has taught more than 200 undergraduates.

EXPERIENCE

Innovative Scientific Solutions, Inc.

Software Engineer II — *Dec 2024 – Present*

- Build full-stack platforms for probabilistic nuclear weapons effects simulation, covering blast, neutron radiation, thermal fluence, ground shock, cratering, and X-ray effects, for defense and allied government agencies.
- Develop user interfaces in Angular, TypeScript, and Tauri, and contribute extensively to UI/UX design.
- Build back-end services in Rust and Python (Flask) with WebSocket communication, MongoDB, and Docker.
- Spearheaded an internal Python benchmarking tool (CLI, then GUI) that runs any combination of the platform's 13 effect calculations and charts the results; it exposed a 10x slowdown in an external-library interface, made the case to leadership with clear graphs, and saved an estimated several weeks of manual testing.
- Port legacy physics models from Fortran and Python to Rust, and design components with UML (StarUML).
- Review merge requests and maintain code quality with GitLab CI/CD and SonarQube on a 4–5 person CUI/ITAR team.

Colorado School of Mines — Golden, CO

Postgraduate Research Assistant, Department of Computer Science — *May 2024 – Dec 2024*

- Developed object detection models for semiconductor cleanroom equipment and integrated them, with Python back-end services, into a Unity AR training app, delivering a working demo in collaboration with the University of Arizona (advisor: Dr. Mehmet Belviranli).

Graduate Research & Teaching Assistant — *Jan 2024 – May 2024*

- Researched graph databases for water treatment plant operations in the early phase of a National Alliance for Water Innovation (NAWI) project (advisor: Dr. Gabe Fierro); assisted with CS instruction.

Tyler Technologies — Lakewood, CO

Software Development Intern — *May 2023 – Aug 2023*

- Migrated a major legacy VB6 desktop application to C# as part of a cloud modernization effort.
- Overhauled a 2,000-test automated suite, restoring about 1,500 tests (75%) to passing.

Colorado School of Mines — Golden, CO

Undergraduate Research Fellow — *Sep 2020 – Apr 2021*

- Built the front end and back end of a collaborative VR environment for engineering education (advisor: Dr. Sebnem Duzgun); co-authored *Collaborative Environment for Education in VR* (Mines research conference, 2021).

TEACHING

Colorado School of Mines — Golden, CO

Adjunct Professor, Department of Computer Science — *Jan 2023 – Dec 2023*

- Co-taught CSCI 220: Data Structures and Algorithms to 120 undergraduates (Fall 2023).
- Taught a section of CSCI 102: Intro to Computer Science Lab to 90 undergraduates (Spring 2023).

Computer Science Outreach Intern & Webmaster — *May 2021 – Aug 2022*

- Designed and taught STEM and computer science summer camps for grades 3–10, co-presented a teacher professional development workshop at InnEdCO 2021, and maintained the DECtech outreach website.

EDUCATION

Colorado School of Mines — Golden, CO

Master of Engineering in Nuclear Engineering (in progress) — *Aug 2026 – Expected May 2031*

Master of Science in Computer Science • GPA 3.97 — *Jan 2022 – May 2024*

Bachelor of Science in Computer Science • Magna Cum Laude • GPA 3.77 — *Aug 2019 – Dec 2022*

Certification: CAE-CDE Certificate, NSA/DHS National Center of Academic Excellence in Cyber Defense Education

Honors: CACI C-MAPP Scholar, National Merit Scholar, Dean's List, Eagle Scout, Rotary Youth Leadership Award

SKILLS

Languages: Rust, Python, TypeScript, JavaScript, C++, C#, Fortran, SQL, HTML/CSS

Frameworks & Tools: Angular, Tauri, Flask, Unity, MongoDB, Graph DBs, Docker, GitLab CI, SonarQube

Domains: Scientific Computing, Object-Oriented Design & UML, UI/UX Design, Computer Vision, Cybersecurity, TDD

Personal Projects: FPGA design (Verilog), 6502 & Z80 assembly programming, CAD/3D printing

Spoken Languages: English (native), Spanish (professional working), German (limited working), Hebrew (elementary)